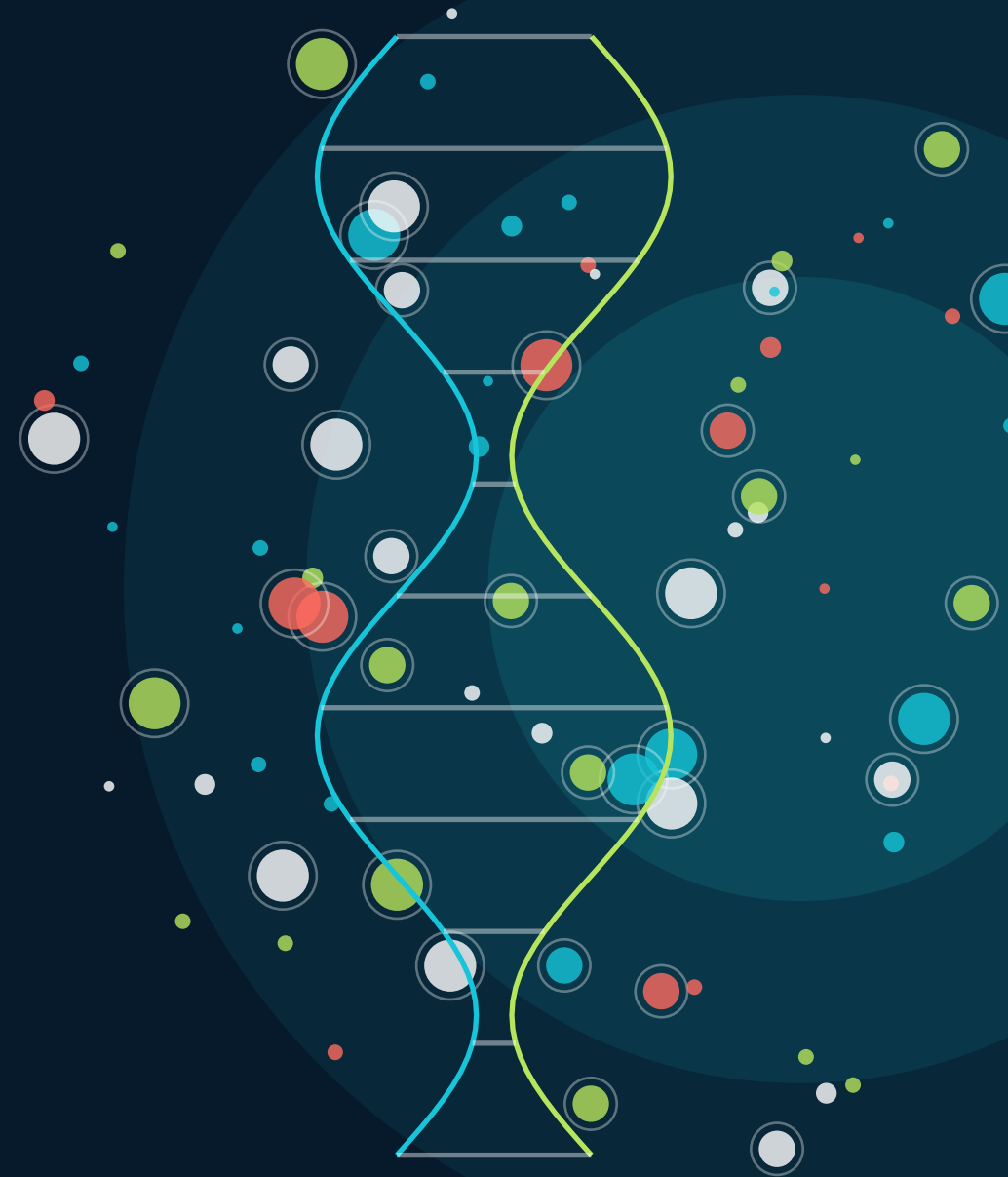


PROFESSIONAL PROGRAM

# ADVANCED MICROBIOLOGY

From microbial systems to genomic intelligence.



01 WET LAB

02 OMICS

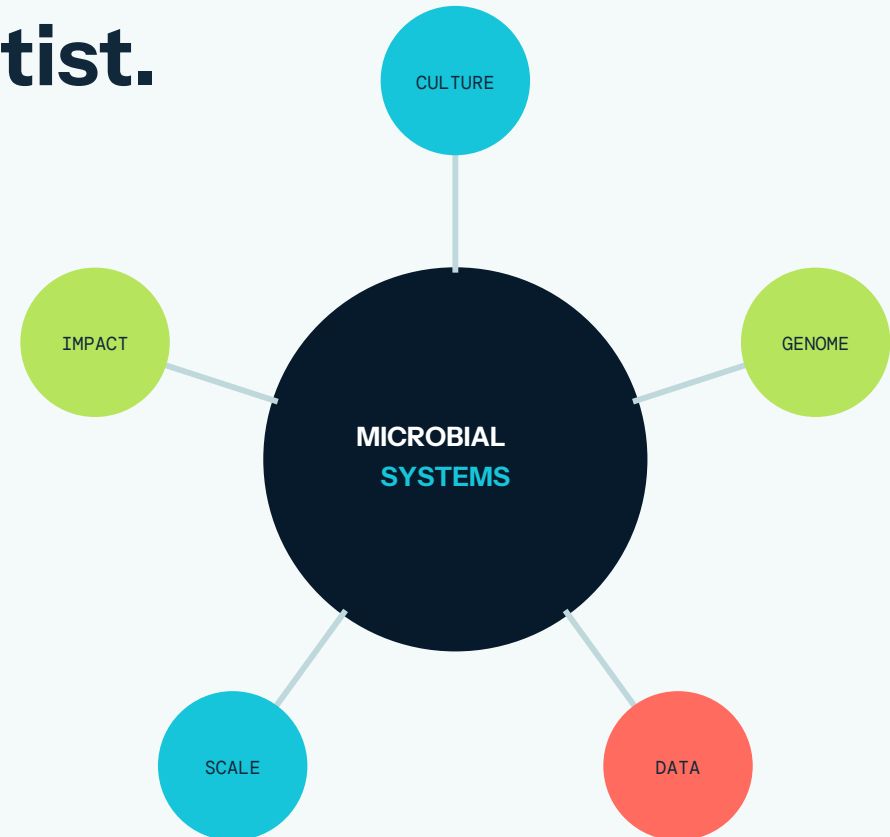
03 BIOPROCESS

LEARN • EXPERIMENT • ANALYSE • TRANSLATE

WHY THIS PROGRAM

# Move beyond observation. Think like a microbial scientist.

A practice-led learning experience connecting classical microbiology with modern molecular, computational and industrial workflows.



**6** CONNECTED MODULES

**3** CAPSTONE PATHWAYS

**1** PORTFOLIO-READY OUTCOME

# Six connected modules

01

## MICROBIAL FOUNDATIONS

Diversity • physiology • metabolism



02

## MOLECULAR TOOLKIT

DNA/RNA • PCR • cloning



03

## GENOMICS + OMICS

Sequencing • annotation •  
metagenomics



04

## CLINICAL SYSTEMS

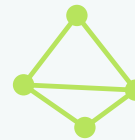
Diagnostics • AMR • immunology



05

## ENVIRONMENT + FOOD

Microbiomes • safety • surveillance



06

## INDUSTRIAL SCALE-UP

Fermentation • QC • downstream  
processing



# The frontier layer

Study the methods reshaping research, healthcare, food systems and biotechnology.

01

## METAGENOMICS

Community profiling and functional insight

02

## CRISPR SYSTEMS

Editing logic, guide design and controls

03

## ANTIMICROBIAL RESISTANCE

Mechanisms, susceptibility and stewardship

04

## BIOINFORMATICS

Sequence QC, alignment and annotation

05

## SYNTHETIC BIOLOGY

Design-build-test-learn thinking

06

## MICROBIOME SCIENCE

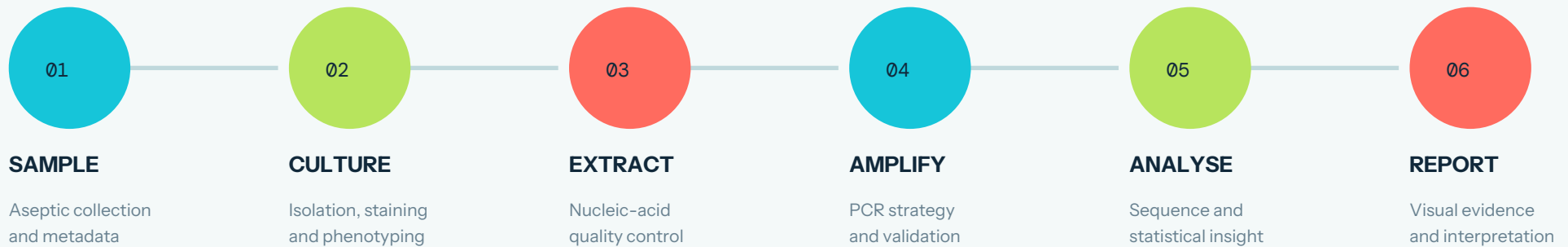
Host, soil and environmental communities



## INTEGRATED WORKFLOW

# From sample to scientific story

Each stage produces evidence, decisions and portfolio-ready documentation.



## QUALITY GATES

## STERILITY



## YIELD



## PURITY



## TRACEABILITY



## APPLIED PROJECTS

# Build evidence, not just notes

Choose a pathway that links experimental design, analysis and scientific communication.



AGRI / ENVIRONMENT

**BIOCONTROL DISCOVERY**

Isolate and screen candidate biopesticide or biofungicide strains.



INDUSTRIAL BIOTECH

**ENZYME + METABOLITE PIPELINE**

Optimise microbial production and evaluate activity, yield and process variables.



CLINICAL / PUBLIC HEALTH

**AMR SURVEILLANCE STUDY**

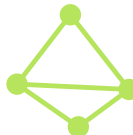
Profile susceptibility patterns and interpret resistance mechanisms responsibly.



GENOMICS / DATA

**MICROBIOME DATA STORY**

Turn community sequence data into taxonomic and functional insight.



MOLECULAR SYSTEMS

**QUORUM SENSING INVESTIGATION**

Explore microbial communication, biofilms and intervention strategies.



GENETIC ENGINEERING

**RECOMBINANT WORKFLOW**

Plan restriction, ligation, transformation and screening with proper controls.

WHAT YOU BUILD

# A professional evidence portfolio

Capture the reasoning behind the result, not only the result itself.

01

## LAB NOTEBOOK

Traceable methods, observations and deviations

02

## DATA PACK

Clean tables, annotated plots and QC decisions

03

## RESEARCH POSTER

A concise visual argument for your findings

04

## PROJECT DEFENCE

A structured explanation of design and interpretation

ASEPTIC TECHNIQUE

PCR DESIGN

SEQUENCE LITERACY

BIOPROCESS THINKING

SCIENTIFIC WRITING

# ACCREDITED CERTIFICATIONS.

Industry-recognised credentials earned across internship, training and capstone delivery.



## INTERNSHIP CERTIFICATE

Verified internship completion credential

01



## INDUSTRIAL TRAINING

Certified practical training hours

02



## PROJECT COMPLETION

Capstone delivery and review proof

03



# GLOBAL MNC CERTIFICATIONS.

Preparation pathways for globally recognised credentials from leading technology organisations.



**CISCO**

**Cybersecurity CCST**



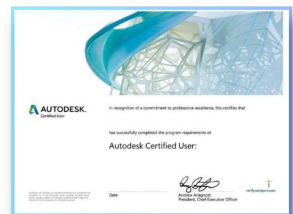
**MICROSOFT**

**Azure Fundamentals**



**IBM**

**Internet of Things**



**AUTODESK**

**Certified User**



**ESB**

**Entrepreneurship**

# AUTHORISED PARTNERS.

An authorised certification and testing network connecting learners with globally accepted credentials.



ADOBE



MICROSOFT



QUICKBOOKS



AUTODESK



CISCO



ESB

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